PANDIYARAJ A

Python Developer | AI/ML Engineer | Full Stack Developer | Data Analyst

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PROFESSIONAL SUMMARY

Innovative MCA student with strong expertise in Python, data analysis, machine learning, NLP, and full-stack development. Proven ability to build AI-powered applications, predictive models, NLP pipelines (NLTK, spaCy, RAG), and scalable web solutions with emphasis on efficiency and clean architecture. Completed a hands-on Data Science internship delivering EDA pipelines, ML classification models, and Power BI dashboards. Passionate about leveraging data and modern AI technologies to solve complex real-world problems.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, SQL
- **Web Development:** Flask, Node.js, HTML, CSS, REST APIs, OOP, Data Structures, Full Stack Development
- **AI, ML & NLP:** CNN, Transfer Learning, Predictive Modeling, NLTK, spaCy, RAG (Retrieval-Augmented Generation), LLMs, Model Evaluation
- **Data Engineering & Database:** Oracle SQL, Database Design, Query Optimization, ETL Pipelines, Pandas, NumPy
- **Data Visualization & Reporting:** Power BI, Excel, Matplotlib, Dashboard Development
- **Tools & Frameworks:** TensorFlow, Keras, OpenCV, Git, GitHub, VS Code, PyCharm, Jupyter Notebook

EXPERIENCE

Data Science Intern *Apr 2025 – Jul 2025*

Edu Tantr, Bengaluru

- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on 10,000+ records using Python (Pandas, NumPy), reducing data inconsistencies by 30%
- Applied feature engineering techniques including encoding, scaling, and handling missing values to optimize ML model performance
- Implemented and evaluated fundamental machine learning algorithms for classification tasks, improving prediction accuracy through hyperparameter tuning
- Executed SQL queries for structured data extraction, transformation, and analysis across relational databases
- Developed interactive Power BI dashboards visualizing KPIs and business insights for stakeholder reporting

PROJECTS

SelfCook – Image-Based Recipe Extraction & Automation System

- Built a full-stack web application integrating Imagga API for food recognition and Edamam API for recipe, ingredient, and nutrition generation serving 100+ lookups per session
- Automated food name extraction using Imagga API, Google Lens integration, and Puppeteer-based web scraping
- Implemented dual search functionality (image + text input) to enhance usability and reduce user friction
- Developed scalable Node.js backend with SQL database, integrating multiple APIs via RESTful architecture

AI-Powered Smart Crop Disease Prediction & Health Monitoring System

- Developed deep learning crop disease detection system using MobileNetV2 (Transfer Learning) for 38-class multi-class classification with high accuracy
- Applied image preprocessing and augmentation (rotation, flipping, zoom) to improve model generalization by 15%
- Designed classification layers using GlobalAveragePooling, Dropout, and Softmax activation in TensorFlow/Keras
- Deployed trained CNN models (.h5) via Flask web application for real-time disease prediction accessible to end users

EDUCATION

Master of Computer Applications (MCA) *Jul 2024 – Apr 2026*

Alagappa University, Karaikudi

Bachelor of Computer Applications (BCA) *Jun 2021 – Apr 2024*

Alagappa Govt. Arts College, Karaikudi

CERTIFICATIONS

- TCS NQT Score: 79.20%
- Machine Learning with Real World Applications – TCS iON
- Data Analytics Program – QSpiders
- Data Science Internship Program – Edu Tantr
- Data Analytics with GenAI Virtual Internship – Deloitte (Forage)

SOFT SKILLS

- Critical Thinking & Problem Solving
- Adaptability & Learning Agility
- Collaboration & Teamwork
- Effective Communication